

Solution Hints

Sol 1) With the given conditions, the cost matrix can be written as:

	C_1	C_2	C_2'	C_3	C_3'	
u						
1	5 60 +	7 -3	7 -3	6 60	6 6	
0	2 80 +	3 5	3 30	5 10	5 35	
-3	0 50	0 55	-M +	0 +	-M +	105
v	3	3	3	5	5	

$$\text{Profit} = 35 + 210 + 225 + 175 + 150 = 795$$

Sol 2) Here total supply is 90 units and total demand is 115 units which shows unbalanced transportation problem. Therefore a dummy row is to be adding, to balance the problem.

After incorporating the conditions given in problem the transportation table will be:

	1	2	3	3'	4	Capacity
A	7	10	14	14	8	30
B	7	11	12	12	6	35
C	5	8	15	15	8	25
Dummy	M	0	0	M	0	53
Demand	20	20	35	28	40	

Applying VAM for getting the IBFS

	1	2	3	3'	4	Capacity	Penalty
A	7	10	14	14 (28)	8 (2)	30	1
B	7	11	12	12	6 (35)	35	1
C	5 (20)	8 (2)	15	15	8 (3)	25	3
Dummy	M	0 (18)	0 (35)	M	0	53	0
Demand	20	20	35	28	40		
Penalty	2	8	12	2	6		

Checking the optimality

	V	-3	0	0	6	0		
U		1	2	3	3'	4	Capacity	Penalty
8	A	(+) 7	(+) 10	(+) 14	14 (28)	8 (2)	30	1
6	B	(+) 7	(+) 11	(+) 12	12	6 (35)	35	1
8	C	5 (20)	8 (2)	(+) 15	15	8 (3)	25	3
0	Dummy	M	0 (18)	0 (35)	M	0	53	0
	Demand	20	20	35	28	40		
	Penalty	2	8	12	2	6		

Optimal solution has reached

$$TC = 14 \times 28 + 8 \times 2 + 6 \times 35 + 5 \times 20 + 8 \times 2 + 8 \times 3 = 758$$

Sol 3)

	Job 1	Job 3	Job 3	Job 4
Machine 1	10	2	3	15
Machine 2	5	10	15	2
Machine 3	15	5	14	7
Machine 4	20	15	13	-

Converting maximization problem into a minimization problem

	Job 1	Job 3	Job 3	Job 4
Machine 1	10	18	17	5
Machine 2	15	10	5	18
Machine 3	5	15	6	13
Machine 4	0	5	7	-

Row reduction

	J1	J3	J3	J4
M 1	5	13	12	0
M 2	10	5	0	13
M 3	0	10	1	8
M 4	0	5	7	-

Column reduction				
	J1	J3	J3	J4
M 1	5	8	12	0
M 2	10	0	0	13
M 3	0	5	1	8
M 4	0	0	7	-

Allocation

M/C	Job
1	4
2	3
3	1
4	2

Total profit of allocation = $(15+15+15+15)*100$
= 6000

Sol 4)

Stationed at B

	10	20	30	40
1	23	24.75	8	11.5
2	20.75	22.5	5.75	9.25
3	15.5	17.25	24.5	4.0
4	14	15.75	23	26.5

Final layover matrix

	10	20	30	40
1	20.5 _A	18.75 _A	8 _B	8 _A
2	20.75 _B	21.0 _A	5.75 _B	9.25 _B
3	4 _A	17.25 _B	19.0 _A	4.0 _B
4	5.5 _A	3.75 _A	20.5 _A	17.0 _A

stationed at A

	1	2	3	4
10	20.5	22.75	4	5.5
20	18.75	21.0	21.75	3.75
30	11.5	13.75	19	20.5
40	8	10.25	15.5	17

Minimizing using hungarian algorithm

	10	20	30	40
1	20.5 _A	18.75 _A	8 _B	8 _A
2	20.75 _B	21.0 _A	5.75 _B	9.25 _B
3	4 _A	17.25 _B	19.0 _A	4.0 _B
4	5.5 _A	3.75 _A	20.5 _A	17.0 _A

Flight	Station	Layover time
1:40	A	8.00
2:30	B	5.75
3:10	A	4.00
4:20	A	3.75

Total layover time = 21.50